

Manoj Chapagain

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SENIOR SOFTWARE ENGINEER

Software Engineer with 8+ years of experience designing and scaling distributed systems that power high-impact products across fintech, cybersecurity, IoT, and SaaS. Known for leading complex projects from concept to production and consistently delivering solutions that enhance reliability, performance, and experience at scale.

WORK EXPERIENCE

Software Engineer — Mechatronic, Canada — *Feb. 2024 – Present*

- Owned full lifecycle of projects from code to production-grade, leveraging Terraform, Kubernetes, and automated CI/CD pipelines
- Built firmware module for resource-constrained IoT devices and responsive React web interface, enabling real-time device management at scale
- Designed, fine-tuned, and deployed AI agents and automation workflows with LangGraph, OpenAI API, and Pinecone, slashing manual ops

Backend Developer — Microsec, Singapore — *May. 2022 – Apr. 2023*

- Engineered PKI security protocols from concept to deployment on IoT/OT devices, using Python Cryptography library for certificate provisioning, encrypted messaging, and integrity
- Architected microservices from design to high-availability clusters, deploying Docker + Kubernetes across multi-node setups with high-availability and fast auto-scaling

Sr. Software Engineer — Codavatar, Kathmandu, Nepal — *Jan. 2021 – Mar. 2022*

- Led backend team from roadmap to on-time delivery of production services, hitting every milestone under tight deadlines
- Optimized Krispcall cloud telephony from core features to 50 k concurrent calls, integrating WebRTC, Twilio, FastAPI, and Redis
- Refactored codebase with clean architecture; applying domain-driven design to cut technical debt and improve API performance

Full Stack Developer — Khalti Digital Wallet, Lalitpur, Nepal — *Nov. 2017 – Dec. 2020*

- Built fintech frontend from zero to 100 k users in 12 months, scaling React app for early-stage startup
- Engineered PCI-compliant payment gateway and digital wallet from backend to live integrations, using Python/Django, React, and AWS (Lambda, RDS, S3) with 50+ bank/telecom APIs

Jr. Full-Stack Developer — Grepsr, Kathmandu, Nepal — *Dec. 2016 – Nov. 2017*

- Developed web-scraping pipelines and internal projects; using microservices, gRPC, Docker, and Kubernetes

EDUCATION

Project Managemnt - IT — Seneca College, Toronto, Canada — *May. 2023 – Dec. 2023*

Bachelor's degree in Computer Engineering —

Tribhuvan University, IOE Pulchowk Campus, Lalitpur, Nepal — *Nov. 2012 – Sep. 2016*

PUBLICATIONS

Effects of Auto-Tuning and Pitch Normalization on Query by Humming —

International Journal of Advanced engineering, June 2020 — *Source*

SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Shell (bash/zsh)

Frameworks & Libraries: FastAPI, Django/DRF, Flask, React, Node.js, gRPC, GraphQL (Ariadne/Graphene), SQLAlchemy, Redis, RabbitMQ, LangGraph, OpenAI API, Pinecone, scikit-learn, Pandas, Selenium

DevOps & Cloud: Kubernetes (K8s), Docker, Terraform, AWS (EC2, S3, RDS, Lambda), CI/CD (GitHub Actions, GitLab CI), Ansible, Nginx, Prometheus, Grafana

Testing & Docs: Pytest, Cypress, Postman, Swagger/OpenAPI, Ruff, Flake8, MkDocs

Others: Linux, Jira, Asana